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Network Tomography Based on Additive Metrics

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Abstract

Inference of the network structure (e.g., routing topology) and dynamics (e.g., link performance) is an essential component in many network design and management tasks. In this paper we propose a new, general framework for analyzing and designing routing topology and link performance inference algorithms using ideas and tools from phylogenetic inference in evolutionary biology. The framework is applicable to a variety of measurement techniques. Based on the framework we introduce and develop several polynomial-time distance-based inference algorithms with provable performance. We provide sufficient conditions for the correctness of the algorithms. We show that the algorithms are consistent (return correct topology and link performance with an increasing sample size) and robust (can tolerate a certain level of measurement errors). In addition, we establish certain optimality properties of the algorithms (i.e., they achieve the optimal l-infinity-radius) and demonstrate their effectiveness via model simulation.

Real-World Impact & Application

The Bridge Between Evolutionary Biology & Internet Mapping: Ni & Tatikonda discovered a profound mathematical duality: the packet loss and delay equations used to map Internet topologies have the exact same additive distance structure as DNA mutation rates used to reconstruct evolutionary trees. This enabled network engineers to repurpose 30+ years of computational biology algorithms directly for Internet monitoring.